

Naresh Medaram

Cloud Lead Network & DevOps Engineer | Hybrid Cloud & Distributed Systems Architecture |

AI Infrastructure Architect

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PROFESSIONAL SUMMARY

Cloud Network Engineer with 14+ years architecting and operating hybrid cloud network infrastructure across on-premises datacenters, GCP, and AWS. Deep L2-L7 networking expertise - routing, switching, load balancing, firewall architecture, and DNS - with a network design philosophy built on non-overlapping address space (NATless) across multi-cloud and on-premises environments, applied consistently for 14+ years and directly extensible to IPv6 and next-generation addressing architectures. Python-centric automation (11,569+ files) underpins a broader infrastructure-as-code practice spanning 81,000+ files in Terraform, Ansible, and Python. Architected PRISM, an 18-microservice distributed incident-response platform reducing mean-time-to-resolution by 97% through event-driven, retry-aware remediation. Builds internal engineering productivity platforms - self-service tooling adopted by 40+ engineers - to reduce operational friction at scale.

CORE EXPERTISE

Hybrid Cloud Network Architecture (GCP, AWS, On-Prem) • Non-Overlapping Address Space (NATless) Design • Infrastructure as Code (Terraform, Ansible, Python) • Distributed Systems & Event-Driven Automation • Kubernetes Networking (CNI, CoreDNS, Service/Ingress) • BGP & Multi-Region Routing • Zero-Trust Network Segmentation

KEY ACHIEVEMENTS

- **AWS Network Integration:** Led integration of 40+ AWS VPCs from a newly acquired environment within a 2-month timeline - redesigned VPC peering, Transit Gateway, Direct Connect, security groups, and route tables; converted to Terraform; consolidated connectivity via Alkira SD-WAN
- **GCP Multi-Region Architecture:** Designed and operated infrastructure across 4 global geo-locations (US, Canada, Europe, Australia)- 50+firewalls,100+ load balancers,and multi-tenant workloads with 99.9% uptime SLA
- **PRISM Distributed Platform:** 18-microservice event-driven incident response platform - MTTR reduced from 2 hours to 72 seconds (97%), 70% fully automated resolution, 22:1 ROI
- **Infrastructure as Code at Scale:** 81,000+ files (Terraform 4,265 / Ansible 10,676 / Python 11,569) developed over 11 years with zero production incidents
- **Zero-Trust Network Segmentation:** 500+ firewall rules across 4 phases, reducing attack surface by 80%+

TECHNICAL SKILLS

AWS Networking: Multi-VPC network architecture, VPC design & peering, Transit Gateway, Direct Connect, security groups, route tables, Alkira SD-WAN

GCP Networking: VPC, Cloud Firewall, Cloud Load Balancing, Cloud DNS, Dedicated/Partner Interconnect, GCVE

Kubernetes Networking: CNI architecture and CIDR allocation strategy, Cilium NetworkPolicy integration, Service/Ingress design, CoreDNS integration with enterprise DNS, eBPF datapath fundamentals

Infrastructure as Code: Terraform,Ansible,Python, GitHub Actions, REST API automation (Palo Alto, F5, GCP, AWS)

Routing & Switching: BGP, OSPF, EIGRP, IS-IS, BFD (fast failure detection), route summarization & aggregation, VLAN/VTP/STP family, VRRP/HSRP/GLBP

Network Security: Zero-trust macrosegmentation, Palo Alto, Illumio microsegmentation, IPsec/SSL VPN, DMVPN

Load Balancing: F5 BIG-IP (LTM/GTM, iRules, iControl), GCP Load Balancing, Alkira

DNS & IPAM: Infoblox, NautoBot, Cloud DNS, Solarwinds

Monitoring & Troubleshooting: Wireshark, tcpdump, SNMP, NetFlow, Syslog, Grafana, Prometheus

Distributed Systems & Workflow Orchestration: Microservices architecture, event-driven automation (Event-Driven Ansible), multi-stage workflow orchestration (GitHub Actions repository & workflow dispatch)

Databases: MySQL, PostgreSQL, pgvector

PROFESSIONAL EXPERIENCE

Cloud Lead Network & DevOps Engineer May 2017 – Present

Ultimate Kronos Group (UKG), Inc.

Hybrid Cloud Network Architecture (GCP + AWS)

- Designed and operated GCP infrastructure across 4 global geo-locations (US, Canada, Europe, Australia) - 50+ firewalls, 100+ load balancers, and multi-tenant workloads with 99.9% uptime SLA

- Led integration of 40+ AWS VPCs from a newly acquired business unit's AWS environment into existing multi-cloud architecture within a 2-month timeline - redesigned VPC peering, Transit Gateway attachments, Direct Connect, security groups, and route tables
- Converted previously unmanaged AWS network configurations into Terraform, unifying the acquired environment with existing GCP infrastructure-as-code
- Replaced complex one-off VPC peering and Transit Gateway topology with Alkira SD-WAN, simplifying cross-environment connectivity and eliminating overlapping configurations (NATless approach)
- Managed full GCVE lifecycle- deployment (2023), operations, and decommission/GCE migration (2026) - with zero downtime
- Established Dedicated and Partner Interconnect with BGP, using global dynamic routing for multi-region reachability from a single connection point

Distributed Systems & Reliability Engineering

- Architected and engineered PRISM in Python - an 18-microservice, event-driven incident response platform built on REST APIs, async diagnostic pipelines, and programmatic remediation workflows
- Designed a 3-phase diagnostic pipeline: parallel live health checks across 8 systems, log correlation analysis, and historical pattern matching for root-cause identification
- Connected diagnostics to Event-Driven Ansible for automated, retry-aware, multi-step remediation workflows
- Result: MTTR reduced from 2 hours to 72 seconds (97% reduction), MTTA reduced from 18 hours to 10 minutes (94% reduction), 70% of incidents resolved with zero human intervention, 22:1 ROI on automation investment
- Eliminated 15+ single points of failure across critical load-balancing infrastructure (100% SPOF elimination) through redundant, multi-zone architecture; validated disaster-recovery readiness through 25+ documented failover tests across multiple data centers

Kubernetes Networking Architecture

- Designed Kubernetes cluster networking architecture - CNI CIDR allocation strategy, Service and Ingress design, CoreDNS integration with enterprise DNS, and Cilium-based NetworkPolicy enforcement - in close collaboration with platform engineering teams
- Designed Kubernetes cluster IP addressing using the 100.64.0.0/10 range, partitioned into per-cluster /16 blocks (supporting up to 256 clusters) with dedicated pod, service, node, and control-plane subnets - eliminating address overlap across hybrid cloud peering and VPN boundaries

Zero-Trust Network Security

- Led enterprise-wide macrosegmentation- 500+ granular firewall rules, reducing attack surface by 80%+
- Deployed Illumio microsegmentation for application-level workload isolation
- Managed 50+ Palo Alto firewalls across 4 global geo-locations with proactive monitoring (Grafana/Prometheus) to prevent CPU exhaustion and session-limit incidents

Infrastructure as Code & Workflow Automation

- Developed and maintained 81,000+ automation files over 11 years (Terraform 4,265 / Ansible 10,676 / Python 11,569) with zero production incidents
- Built subnet provisioning automation reducing turnaround from 3 days to 10 minutes (99% faster) across 320 subnets in 10 regions
- Architected multi-stage, cross-repository workflow orchestration using GitHub Actions repository_dispatch - coordinating dependent automation pipelines with reuse-first resource allocation
- Built retry-aware, event-driven remediation workflows with Event-Driven Ansible, translating monitoring signals into automated multi-step diagnostic and remediation sequences

Engineering Productivity Platforms

- Built an internal self-service chatbot (retrieval-augmented) that reduced network documentation search time by 60% and was adopted by 40+ engineers, reducing dependency on tribal knowledge
- Engineered a Python REST API service exposing 13 firewall-intelligence tools at sub-100ms latency - natural-language query interface integrated into engineers' developer tooling, reducing manual firewall analysis from 30-60 minutes to instant lookups

Production Network Operations

- Performed production network troubleshooting using packet-level analysis (Wireshark, tcpdump) across multi-vendor and containerized (GKE) environments
- Managed DNS, DHCP, and IPAM via Infoblox and NautoBot, consolidating three legacy IPAM systems into a single API-driven source of truth
- Built network observability and telemetry pipelines - SNMP, NetFlow, and Syslog data feeding Grafana and Prometheus dashboards across 50+ firewalls - for proactive incident detection and operational visibility
- Led root-cause analysis for a production incident where a batch data-ingestion pipeline triggered a DNS traffic spike, exhausting firewall session capacity and causing authentication failures; implemented proactive session and CPU monitoring that prevented recurrence

Cross-Team Leadership

- Led 125+ technical initiatives across US and India engineering teams, consistently driving multi-year programs to completion
- Mentored 40+ engineers through self-service platforms and documentation

Cloud Network Engineer*July 2015 – May 2017*

Comtek Global, Inc.

- Designed and deployed multi-vendor network infrastructure (Cisco, Juniper, F5, Palo Alto) - routers, switches, firewalls, load balancers
- Configured static and dynamic routing (BGP, OSPF): Established site-to-site VPN tunnels Juniper & Cisco
- Established WAN connectivity between datacenters via BGP peering with service providers
- Deployed F5 LTM/GTM - virtual servers, pools, custom iRules, load-balancing algorithms
- Managed SSL certificate lifecycle (DigiCert, GoDaddy, Comodo) for customer-facing applications

Graduate/Teaching Assistant*August 2014 – May 2015*

University of Houston - Clear Lake, TX

- Supported 45 graduate students in network engineering coursework and program accreditation
- Created Methods of Procedure (MOPs) for OS upgrades and Layer 2/3 protocol configuration
- Managed Cisco Prime, Infoblox (DNS/DHCP/IPAM), and Cisco ASA firewalls

Network Engineer*June 2013 – November 2013*

Devblaze Pvt. Ltd.

- Deployed Cisco Nexus, ASR, and Catalyst switching/routing infrastructure, including Nexus 2000 FEX
- Configured OSPF, EIGRP, and BGP routing across multi-vendor LAN/WAN infrastructure
- Configured F5 LTM/GTM pools, virtual servers, and custom iRules
- Migrated IPsec VPN tunnels from pre-shared key to certificate-based authentication for scalability

Network Engineer*May 2012 – April 2013*

Levonsys Pvt. Ltd.

- Led Cisco-to-Juniper migration: configuration analysis, conversion, MOPs, and redundancy testing prior to cutover
- Configured switching (RSTP/VSTP/MSTP) and Layer 3 protocols (static, OSPF, EIGRP) on Juniper devices
- Migrated firewall filters and security policies from Cisco to Juniper, validated with senior engineers

Network Administrator*May 2010 – April 2012*

Global Research Solutions

- Configured VLANs, OSPF, and static routing on Cisco Catalyst switches
- Implemented Layer 2 security (BPDU Guard, Root Guard, port security, DHCP attack prevention)
- Performed root-cause analysis for Layer 1-3 issues and managed DNS/DHCP administration

PROFESSIONAL DEVELOPMENT & INDUSTRY ENGAGEMENT

- **Google Cloud Next 2025 - Global AI & Cloud Conference (2025):** Engaged with Google Cloud product and engineering leaders on generative AI infrastructure, multi-cloud architecture, and enterprise AI adoption patterns
- **ONUG AI Summit, New York, NY (2024):** Participated in the Open Networking User Group summit focused on AI-driven network operations and open networking standards for enterprise IT
- **FLNUG AI Summit, Tampa, FL (March 2025):** Invited by Network to Code to serve as a panelist on applied AI for network engineering, automation, and operations - sharing perspectives alongside industry peers
- **Juniper Networks Industry Event, Miami, FL (2025):** Engaged with Juniper Networks roadmap sessions on AI-driven networking, automation, and enterprise security

EDUCATION

Master of Science, Computer Engineering - University of Houston-Clear Lake, Houston, Texas | 2015 | GPA 3.7/4.0

Bachelor of Engineering, Electronics and Communication Engineering

DRK Institute of Science and Technology, Andhra Pradesh, India | 2013 | GPA 81.83%

CERTIFICATIONS

- GCP Professional Cloud Network Engineer | GCP Cloud Digital Leader | GCP Gen AI Leader
- Terraform Associate (HashiCorp) | Claude Code Architect | Langchain Foundation - Python
- Palo Alto Certified Network Security Engineer (PCNSE)
- Cisco Certified Network Professional (CCNP)
- Juniper Networks: JNCIP-ENT, JNCIS-ENT, JNCIA
- F5 Certified Administrator, BIG-IP | Red Hat Certified Specialist in Ansible Automation (EX407) | OpenStack Certified Administrator (COA)